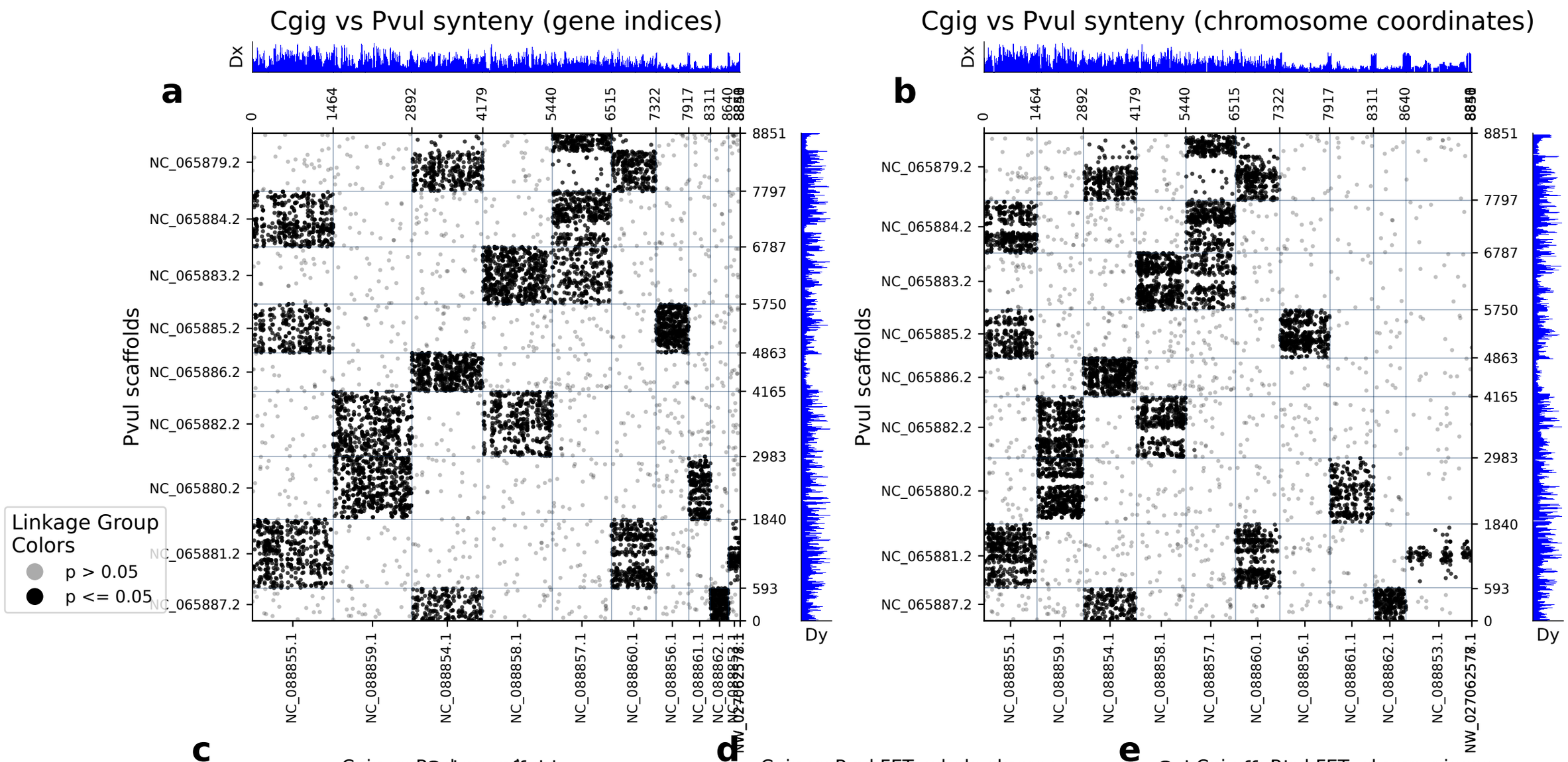
diamond: 8852 orthologs on 12 Cgig scaffolds and 9 Pvul scaffolds.



c

Cgig vs Pvul groups

d

Cgig vs Pvul FET, whole chroms.

	Cgig scaf	Pvul scaf	p (FET)	Count
0	NC_088859.1	NC_065880.2	7.80e-309	690
1	NC_088858.1	NC_065883.2	0.00e+00	668
2	NC_088859.1	NC_065882.2	7.81e-195	591
3	NC_088854.1	NC_065886.2	0.00e+00	576
4	NC_088855.1	NC_065881.2	6.08e-135	544
5	NC_088855.1	NC_065884.2	1.49e-137	484
6	NC_088856.1	NC_065885.2	0.00e+00	474
7	NC_088858.1	NC_065882.2	5.82e-108	449
8	NC_088860.1	NC_065881.2	2.95e-163	420
9	NC_088857.1	NC_065884.2	2.27e-132	405
10	NC_088854.1	NC_065879.2	2.13e-59	350
11	NC_088861.1	NC_065880.2	1.43e-214	314
12	NC_088860.1	NC_065879.2	1.15e-84	299
13	NC_088855.1	NC_065885.2	4.39e-35	292
14	NC_088862.1	NC_065887.2	7.46e-289	274
15	NC_088857.1	NC_065879.2	2.54e-37	272
16	NC_088857.1	NC_065883.2	4.13e-32	258
17	NC_088854.1	NC_065887.2	5.03e-43	219
18	NC_088853.1	NC_065881.2	3.48e-60	132

e

Cgig vs Pvul FET, chrom. pieces

	Cgig scaf	Pvul scaf	p (FET)	Count
0	NC_088859.1 : 0-end	NC_065880.2 : 0-end	7.80e-309	690
1	NC_088858.1 : 0-end	NC_065883.2 : 0-end	0.00e+00	668
2	NC_088859.1 : 0-end	NC_065882.2 : 0-end	7.81e-195	591
3	NC_088854.1 : 0-end	NC_065886.2 : 0-end	0.00e+00	576
4	NC_088855.1 : 0-end	NC_065881.2 : 0-end	6.08e-135	544
5	NC_088855.1 : 0-end	NC_065884.2 : 0-end	1.49e-137	484
6	NC_088856.1 : 0-end	NC_065885.2 : 0-end	0.00e+00	474
7	NC_088858.1 : 0-end	NC_065882.2 : 0-end	5.82e-108	449
8	NC_088860.1 : 0-end	NC_065881.2 : 0-end	2.95e-163	420
9	NC_088857.1 : 0-end	NC_065884.2 : 0-end	2.27e-132	405
10	NC_088854.1 : 0-end	NC_065879.2 : 0-end	2.13e-59	350
11	NC_088861.1 : 0-end	NC_065880.2 : 0-end	1.43e-214	314
12	NC_088860.1 : 0-end	NC_065879.2 : 0-end	1.15e-84	299
13	NC_088855.1 : 0-end	NC_065885.2 : 0-end	4.39e-35	292
14	NC_088862.1 : 0-end	NC_065887.2 : 0-end	7.46e-289	274
15	NC_088857.1 : 0-end	NC_065879.2 : 0-end	2.54e-37	272
16	NC_088857.1 : 0-end	NC_065883.2 : 0-end	4.13e-32	258
17	NC_088854.1 : 0-end	NC_065887.2 : 0-end	5.03e-43	219
18	NC_088853.1 : 0-end	NC_065881.2 : 0-end	3.48e-60	132

(a) depicts the chromosomes/scaffolds of Cgig (x-axis) plotted against the chromosomes/scaffolds of Pvul (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 8852 orthologs found between 12 Cgig scaffolds and 9 Pvul scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chormosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.